# Supporting Information

## Semi-Supervised Deep Learning based Multi-Component Spectral Calibration Modeling for UV-Vis and Near-Infrared Spectroscopy without Information Loss

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## Text S1 Data description

### Diesel Fuels datasets

The diesel dataset is publicly available and can be downloaded at http:/www.eigenvector.com/data/SWRI/in-dex.html. The NIR spectrum measured by the American Society of Testing and Materials (ASTM) ranges from 750-1550 nm with each spectral interval of 2 nm and contains 401 points. The six properties of interest are total aromatics (Total), cetane number (CN), density (D4052), viscosity (Visc), boiling points (BP50), and freezing temperature (Freeze). The data were divided into three groups: t_sd_hl, t_sd_lla, and t_sd_llb, which contains 256 samples. According to the suggestions of the database, this paper combines " t_sd_hl " and " t_sd_lla " to form the training set (138 samples in total) and " t_sd_llb " as the test set with 118 samplewss. The spectrum of diesel data is shown in Figure 5(a).

### Corn datasets

The dataset can be downloaded from the website http:/www.eigenvector.com/data/Corn/index.html. The corn spectral dataset measured by m5 NIR spectrometers contains 80 samples with a wavelength range of 1100-2498nm and 700 wavelength points. The three properties measured were oil, protein, moisture, and starch. The spectrum of corn data is shown in Figure 5(b). The samples were divided into training sets with 60 samples and test sets with 20 samples.

### Multi-metal ions datasets

The dataset of multi-metal ions was obtained by the UV-visible spectrophotometer (Shimadzu Co., Ltd., Japan) with the scanning range of 400-800 nm and the scanning resolution of 1 nm. The reagents are nitroso-R salt (0.40%) as the chromogenic reagent, acetic acid-sodium acetate (pH=5.5) as the buffer solution, cetyltrimethylammonium bromide (CTMAB. 0.01 mol $L^{-1}$) as sensitizer. The main standard solutions were $Zn^{2+}$ standard solution (2.6 mol $L^{-1}$), $Cu^{2+}$ standard solution (0.001 mol $L^{-1}$), $Co^{2+}$ standard solution (0.001 mol $L^{-1}$), $Ni^{2+}$ standard solution (0.001 mol $L^{-1}$) and $Fe^{2+}$ standard solution (0.001 mol $L^{-1}$). All reagents are of analytical grade.

The experimental procedure is as follows: first, add appropriate amounts of $Zn^{2+}$, $Cu^{2+}$, $Co^{2+}$, $Ni^{2+}$, $Fe^{3+}$ standard solutions in a 25 ml volumetric flask. Then add 5.0 ml of acetic acid-sodium acetate buffer and 3.0 ml of CTMAB solution and shake well. Add 2.5 ml of nitroso-R salt. Finally, add distilled water to make up to the volume of 25 ml. The UV-visible absorption spectra of the 115 groups of mixed solutions are measured in the range of 400-800 nm. The original spectral absorbance curves are shown in Figure 5(c).

The 115 sets of labeled samples were divided using the Kennard-Stone (KS) algorithm [1], 86 sets of labeled samples were used as the training set and the remaining 29 sets were used as the test set. To compare the performance of different models, these 86 sets of samples are used for all model training.

## Table S1 Hyper-parameters for modeling

| Parameters | Range |
| --- | --- |
| Optimizer | Adam |
| Activation function | ReLU |
| Batch size | 30-100 |
| Epoch | 100-1000 |
| Original dimension | Number of spectral variables in each dataset |

The optimizer was set as Adam optimizer, which is generally regarded as the best choice for spectral analysis [2]. The activation function is set as a linear rectifier function (ReLU). All the spectral data is firstly normalized by Eq. (S18) to ensure that each variable has a range of [0,1].

$$\boldsymbol{X}_{norm} = \frac{\boldsymbol{x} - \boldsymbol{x}_{\min}}{\boldsymbol{x}_{\max} - \boldsymbol{x}_{\min}} \tag{S1}$$

where $x$ is the original spectral data. The normalized spectral data $X_{norm}$ is then transformed into the corresponding potential variables by the NICEM model which satisfies the Gaussian distribution. The random gradient descent method is used to update the weights until the model convergence. All model parameters are optimized to improve the performance of the model.

## Text S2 Evaluation metrics

The root mean square error of calibration set $RMSEC$, cross-validation set $RMSECV$, test set $RMSEP$ and coefficient of determination $R^2$ are used to assess the performance of each approach.

The root mean square error of calibration set $RMSEC$ is defined as

$$RMSEC = \sqrt{\frac{\sum_{i=1}^{N_{\mathrm{cal}}}(y_{ci} - \hat{y}_{ci})^2}{N_{cal}}} \tag{S2}$$

where $N_{cal}$ is the number of samples in the test set. $y_{ci}$ and $\hat{y}_{ci}$ are the true value and predicted value of the $i$-th samples in the calibration set, respectively.

The root mean square error of cross-validation set $RMSECV$ is defined as

$$RMSECV = \sqrt{\frac{\sum_{i=1}^{N_{\mathrm{cv}}}(y_{vi} - \hat{y}_{vi})^2}{N_{cv}}} \tag{S3}$$

where $N_{cv}$ is the number of samples in the test set. $y_{vi}$ and $\hat{y}_{vi}$ are the true value and predicted value of the $i$-th samples in the cross-validation set set, respectively.

The formulas of $RMSEP$ are illustrated as

$$RMSEP = \sqrt{\frac{\sum_{i=1}^{N_{test}}(y_{ti} - \hat{y}_{ti})^2}{N_{test}}} \tag{S4}$$

where $N_{test}$ is the number of samples in the test set. $y_{ti}$ and $\hat{y}_{ti}$ are the true value and predicted value of the $i$-th samples in the test set, respectively.

The $R^2$ is defined as

$$R^2 = 1 - \frac{\sum_{i=1}^{N_{test}}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{N_{test}}(y_i - \overline{y})^2} \tag{S5}$$

where $\overline{y}$ is the average of the predicted value.

## Text S3 Optimization and evaluation of the comparison methods

In all methods, the optimal number of selected latent variables of PLS model was determined by the minimum mean square prediction error of determination using 10-fold cross-validation in a range of 5-20. For MCUVE and CARS, the number of Monte Carlo sampling runs was set from range [50,100, 200, 500]. For GA-PLS, the probability of cross-over was set to 0.5, and the probability of mutation was set to 0.001. For SVR, the kernel function was set as the radial basis function (RBF). The optimal penalty was set from range [0.001, 0.1, 1, 10, 100, 1000]. For ANN, the activation function was set as relu and the solver was set as adam. The initial learning rate was set as 0.001, and the maximum number of iterations was set from range 500 to 2000. For ELM, the activation function was set as sigmoidal, and the number of neurons was set from 20 to 100.

**Table S2 The comparison of generalized performance between NICEM method and different variable selection-based methods for diesel fuels datasets. The best model for each component is marked in bold**

| Content | Method | VARs[a] | LVs[b] | RMSECV | RMSEC | RMSEP | $R^2$ |
|---|---|---|---|---|---|---|---|
| BP50 | PLS | 401 | 10 | 4.496 | 4.272 | 4.876 | 0.893 |
| | MCUVE | 148 | 11 | 4.134 | 4.043 | 4.452 | 0.91 |
| | CARS | 25 | 9 | 4.27 | 4.11 | 4.575 | 0.907 |
| | GA-PLS | 58 | 11 | 3.921 | 3.815 | 4.36 | 0.928 |
| | NICE | 401 | 12 | 2.842 | 2.785 | 3.172 | 0.964 |
| | **NICEM** | **130** | **9** | **2.705** | **2.617** | **2.953** | **0.969** |
| CN | PLS | 401 | 7 | 1.808 | 1.712 | 2.077 | 0.638 |
| | MCUVE | 108 | 7 | 1.769 | 1.694 | 2.056 | 0.645 |
| | CARS | 43 | 7 | 1.848 | 1.798 | 2.103 | 0.626 |
| | GA-PLS | 70 | 8 | 1.857 | 1.784 | 2.036 | 0.653 |
| | NICE | 401 | 8 | 1.771 | 1.702 | 1.903 | 0.673 |
| | **NICEM** | **155** | **8** | **1.712** | **1.652** | **1.832** | **0.697** |
| D4052 | PLS | 401 | 18 | 0.00115 | 0.00109 | 0.00122 | 0.985 |
| | MCUVE | 107 | 10 | 0.00112 | 0.00106 | 0.00118 | 0.986 |
| | CARS | 61 | 9 | 0.00101 | 0.00098 | 0.00109 | 0.989 |
| | GA-PLS | 79 | 10 | 0.00104 | 0.00102 | 0.00115 | 0.987 |
| | NICE | 401 | 16 | 0.00087 | 0.00081 | 0.00095 | 0.991 |
| | **NICEM** | **210** | **15** | **0.00076** | **0.00071** | **0.00082** | **0.993** |
| Freeze | PLS | 401 | 7 | 2.606 | 2.463 | 2.726 | 0.529 |
| | MCUVE | 78 | 8 | 2.542 | 2.472 | 2.713 | 0.533 |
| | CARS | 45 | 7 | 2.557 | 2.502 | 2.752 | 0.519 |
| | GA-PLS | 56 | 8 | 2.451 | 2.348 | 2.577 | 0.579 |
| | NICE | 401 | 10 | 2.404 | 2.32 | 2.551 | 0.587 |
| | **NICEM** | **75** | **9** | **2.358** | **2.241** | **2.463** | **0.615** |
| Total | PLS | 401 | 10 | 0.676 | 0.627 | 0.728 | 0.985 |
| | MCUVE | 114 | 9 | 0.679 | 0.635 | 0.741 | 0.984 |
| | CARS | 52 | 10 | 0.752 | 0.724 | 0.783 | 0.982 |
| | GA-PLS | 85 | 9 | 0.762 | 0.738 | 0.816 | 0.981 |
| | NICE | 401 | 12 | 0.553 | 0.523 | 0.572 | 0.992 |
| | **NICEM** | **390** | **9** | **0.545** | **0.506** | **0.561** | **0.993** |
| Visc | PLS | 401 | 12 | 0.094 | 0.082 | 0.101 | 0.932 |
| | MCUVE | 76 | 10 | 0.098 | 0.084 | 0.108 | 0.928 |
| | CARS | 53 | 11 | 0.086 | 0.075 | 0.099 | 0.931 |
| | GA-PLS | 68 | 9 | 0.083 | 0.071 | 0.096 | 0.933 |
| | NICE | 401 | 10 | 0.074 | 0.064 | 0.086 | 0.95 |
| | **NICEM** | **225** | **14** | **0.073** | **0.06** | **0.081** | **0.956** |

[a] *VARs: The number of selected variables.*

[b] *LVs: The number of selected latent variables of PLS.*

**Figure S1 Relationship between RMSECV and number of selected latent variables (LVs) in PLS and NICEM models for diesel fuels datasets. (a) BP50, (b) CN, (c) D4052, (d) Freeze, (e) Total, (f) Visc. The blue dashed line represents optimal latent variables for PLS. The red dashed line represents the optimal latent variables for NICEM.**

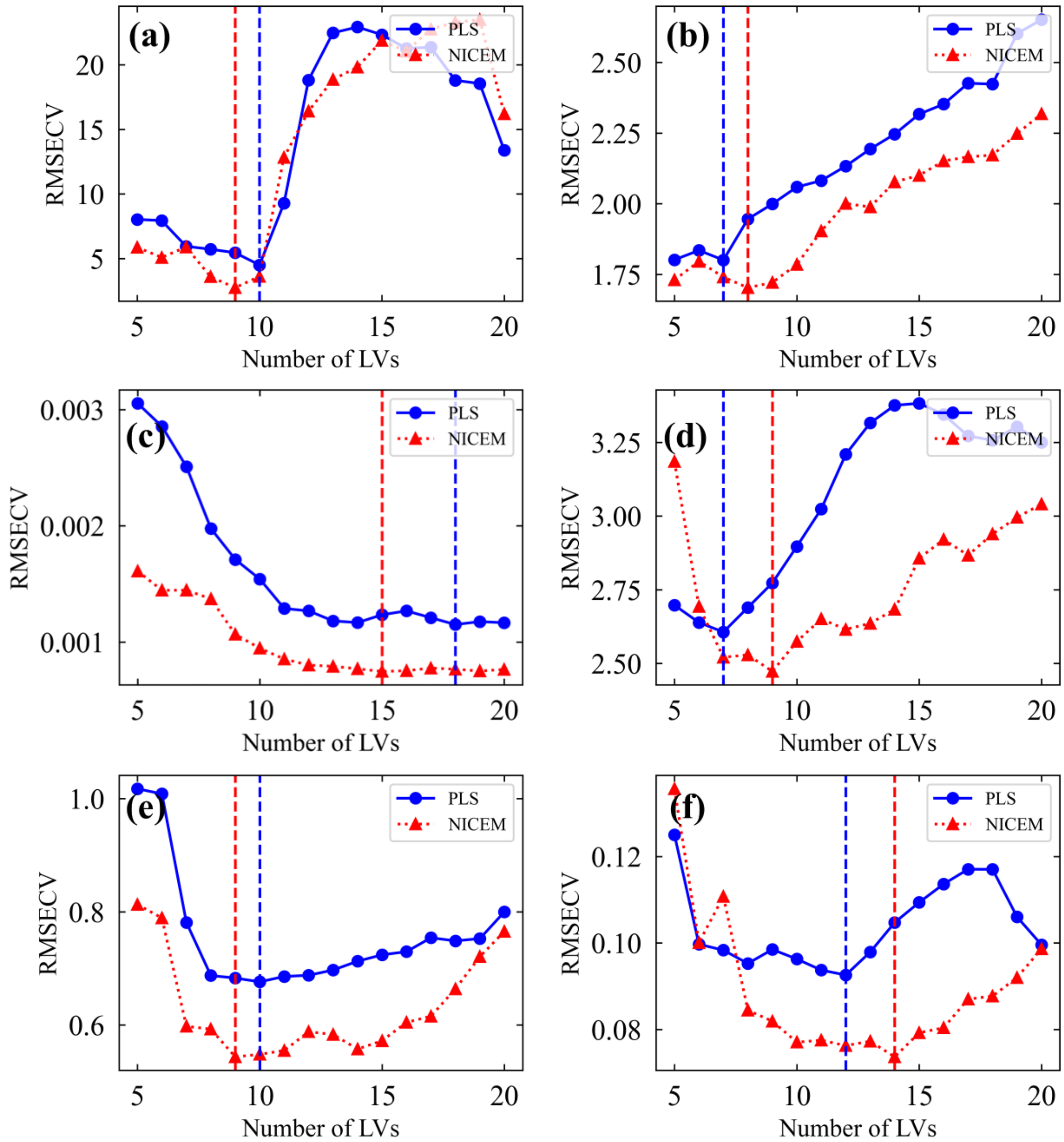
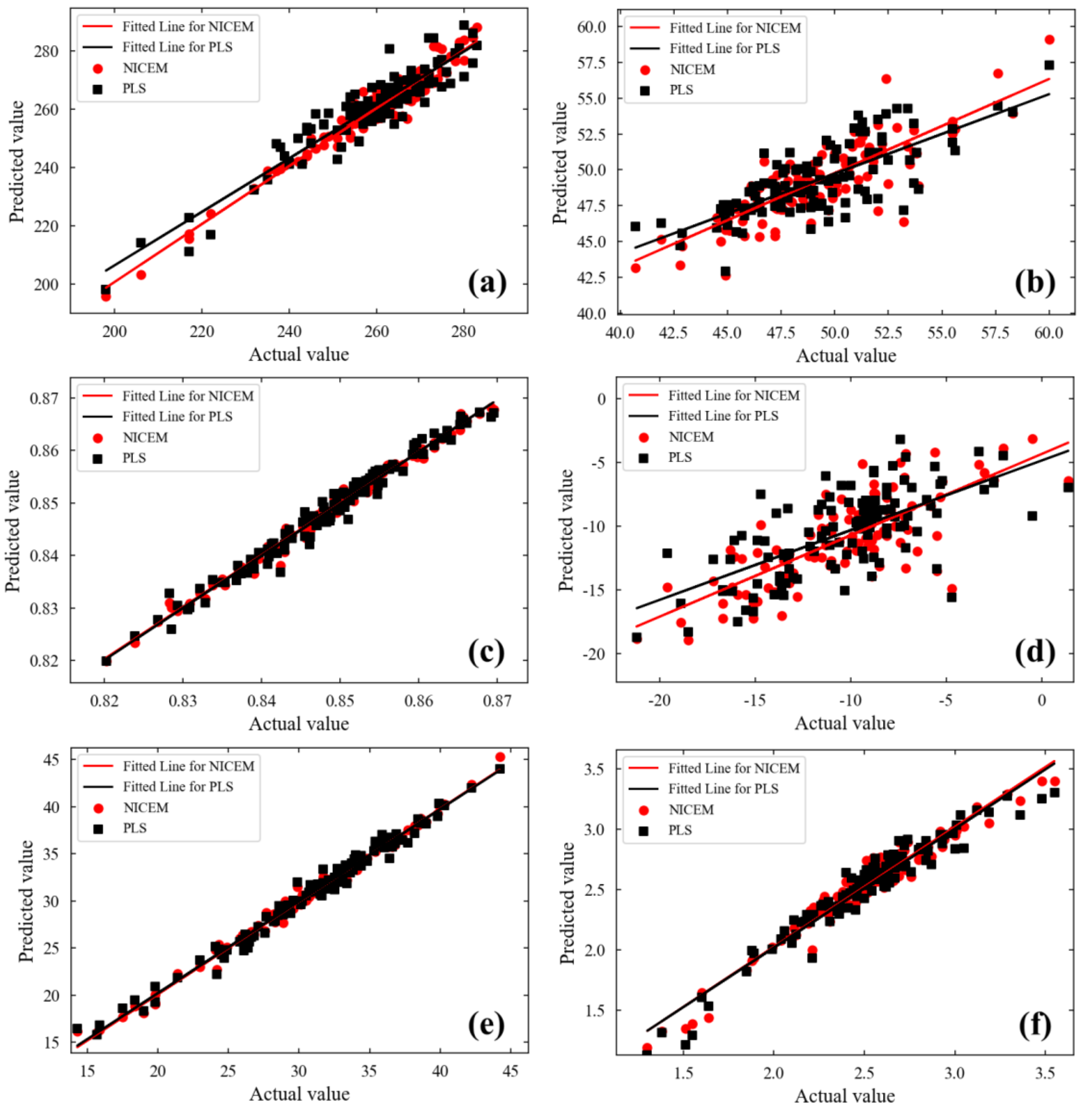


**Figure S2 Prediction results of PLS and NICEM models for diesel fuels datasets. (a) BP50, (b) CN, (c) D4052, (d) Freeze, (e) Total, (f) Visc.**



## Table S3 Prediction results of previous studies for diesel fuels datasets

| Datasets | Content | Calibration Method | RMSEP | $R^2$ | Reference |
|---|---|---|---|---|---|
| Diesel fuels dataset | D4052 | FOSGD–SCARS | 1.78 | / | 3 |
| | CN | FOSGD–SCARS | 2.33 | / | 3 |
| Diesel fuels dataset | BP50 | BOSS | 3.286 | 0.9630 | 4 |
| | CN | BOSS | 2.0844 | 0.6497 | 4 |
| | Freeze | BOSS | 2.6345 | 0.5597 | 4 |
| | Total | BOSS | 0.6138 | 0.9897 | 4 |
| | Visc | BOSS | 0.0951 | 0.9401 | 4 |
| Diesel fuels dataset | Total | VCPA-IRIV | 0.6004 | 0.9901 | 5 |
| | | BOSS-IRVS | 0.6026 | 0.9903 | 5 |

**Table S4 The comparison of generalized performance between NICEM method and different variable selection-based methods for corn datasets**

| Content | Method | VARs | LVs | RMSECV | RMSEC | RMSEP | $R^2$ |
|---|---|---|---|---|---|---|---|
| Moisture | PLS | 700 | 7 | 0.058 | 0.055 | 0.071 | 0.964 |
| | MCUVE | 37 | 8 | 0.059 | 0.057 | 0.077 | 0.958 |
| | CARS | 10 | 5 | 0.065 | 0.061 | 0.079 | 0.956 |
| | GA-PLS | 27 | 7 | 0.06 | 0.056 | 0.067 | 0.968 |
| | NICE | 700 | 12 | 0.042 | 0.037 | 0.045 | 0.974 |
| | **NICEM** | **95** | **11** | **0.033** | **0.032** | **0.044** | **0.985** |
| Oil | PLS | 700 | 10 | 0.061 | 0.051 | 0.064 | 0.886 |
| | MCUVE | 41 | 8 | 0.064 | 0.06 | 0.089 | 0.773 |
| | CARS | 25 | 7 | 0.058 | 0.048 | 0.061 | 0.895 |
| | GA-PLS | 32 | 7 | 0.051 | 0.042 | 0.06 | 0.897 |
| | NICE | 700 | 11 | 0.043 | 0.033 | 0.051 | 0.904 |
| | **NICEM** | **89** | **7** | **0.016** | **0.012** | **0.019** | **0.953** |
| Protein | PLS | 700 | 10 | 0.129 | 0.119 | 0.142 | 0.927 |
| | MCUVE | 71 | 8 | 0.136 | 0.125 | 0.143 | 0.925 |
| | CARS | 22 | 7 | 0.12 | 0.112 | 0.148 | 0.919 |
| | GA-PLS | 47 | 7 | 0.117 | 0.101 | 0.138 | 0.931 |
| | NICE | 700 | 10 | 0.102 | 0.089 | 0.115 | 0.958 |
| | **NICEM** | **46** | **8** | **0.068** | **0.065** | **0.077** | **0.981** |
| Starch | PLS | 700 | 7 | 0.443 | 0.424 | 0.504 | 0.587 |
| | MCUVE | 86 | 9 | 0.438 | 0.413 | 0.491 | 0.604 |
| | CARS | 54 | 8 | 0.417 | 0.396 | 0.472 | 0.636 |
| | GA-PLS | 49 | 9 | 0.388 | 0.361 | 0.459 | 0.658 |
| | NICE | 700 | 11 | 0.129 | 0.108 | 0.148 | 0.93 |
| | **NICEM** | **39** | **10** | **0.092** | **0.086** | **0.113** | **0.979** |

**Figure S3 Relationship between RMSECV and number of selected latent variables (LVs) in PLS and NICEM models for corn data set. (a) Moisture, (b) Oil, (c) Protein, (d) Starch.**

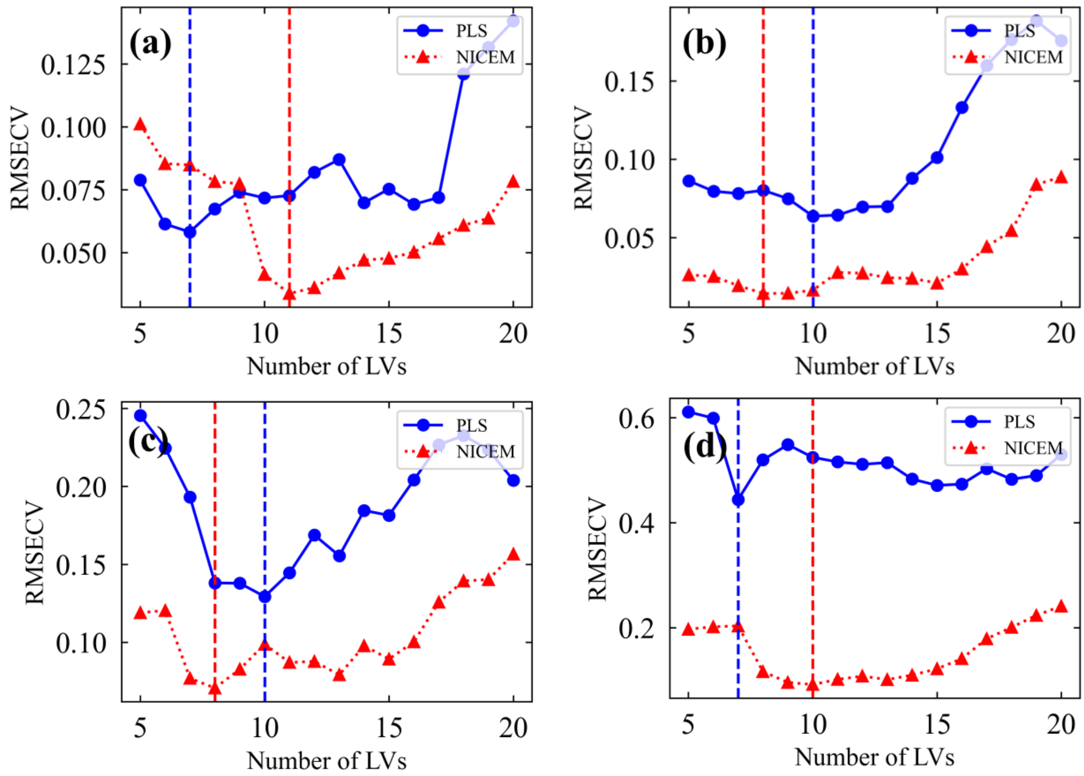
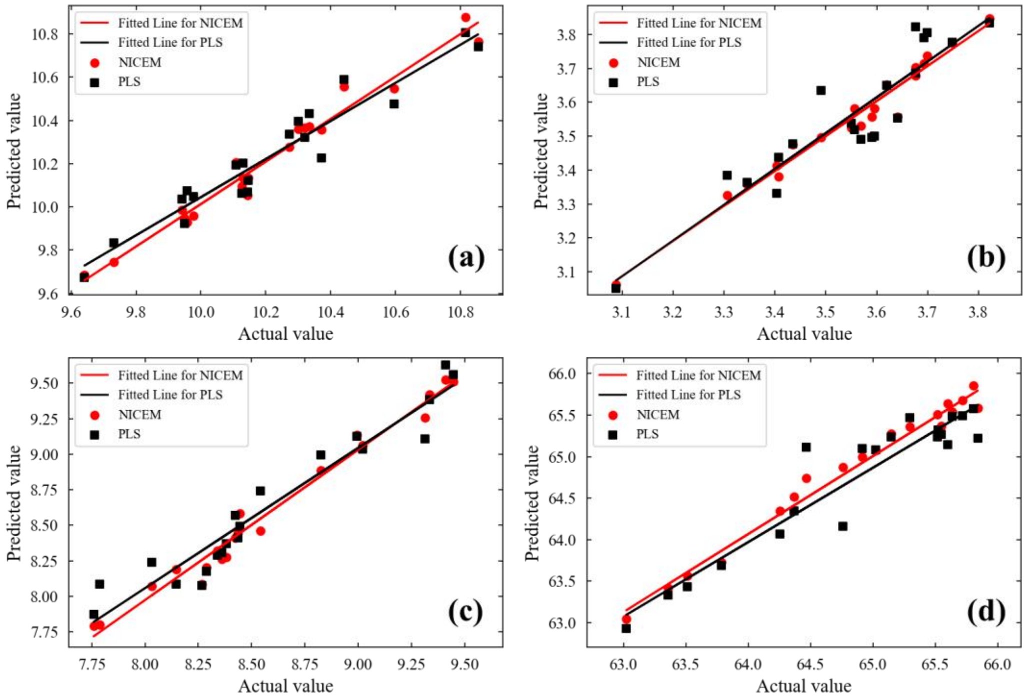


**Figure S4 Prediction results of PLS and NICEM models for corn datasets. (a) Moisture, (b) Oil, (c) Protein, (d) Starch.**



## Table S5 Prediction results of previous studies for corn datasets

| Datasets | Content | Calibration Method | RMSEP | $R^2$ | Reference |
|---|---|---|---|---|---|
| Corn dataset | protein | DeepSpectra | 0.12 | 0.91 | 6 |
| Corn dataset | Moisture | ECNN | / | 0.9471 | 7 |
| | Oil | ECNN | / | 0.8079 | 7 |
| | Protein | ECNN | / | 0.8172 | 7 |
| | Starch | ECNN | / | 0.7278 | 7 |
| Corn dataset | Moisture | BP ANN + L1 | / | 0.9252 | 7 |
| | Oil | BP ANN + L1 | / | 0.7121 | 7 |
| | Protein | BP ANN + L1 | / | 0.8005 | 7 |
| | Starch | BP ANN + L1 | / | 0.8732 | 7 |
| Corn dataset | Starch | DeepSpectra | 0.18 | / | 8 |
| | Starch | M3GPSpectra | 0.16 | / | 8 |

**Table S6 The comparison of generalized performance between NICEM method and different variable selection-based methods for multi-metal ions datasets**

| Content | Method | VARs | LVs | RMSECV | RMSEC | RMSEP | $R^2$ |
|---|---|---|---|---|---|---|---|
| Zinc | PLS | 400 | 9 | 0.4754 | 0.4503 | 0.4815 | 0.7276 |
| | MCUVE | 125 | 11 | 0.4073 | 0.3963 | 0.4124 | 0.8001 |
| | CARS | 78 | 8 | 0.4079 | 0.3846 | 0.4019 | 0.8102 |
| | GA-PLS | 84 | 10 | 0.4245 | 0.4138 | 0.4479 | 0.7643 |
| | NICE | 400 | 14 | 0.3443 | 0.3351 | 0.3567 | 0.8504 |
| | **NICEM** | **65** | **12** | **0.274** | **0.2634** | **0.2887** | **0.9027** |
| Copper | PLS | 400 | 14 | 0.1378 | 0.1318 | 0.1463 | 0.9292 |
| | MCUVE | 114 | 7 | 0.1267 | 0.1224 | 0.1399 | 0.9352 |
| | CARS | 59 | 8 | 0.1075 | 0.1079 | 0.1126 | 0.9581 |
| | GA-PLS | 75 | 10 | 0.1094 | 0.1043 | 0.1125 | 0.9516 |
| | NICE | 400 | 16 | 0.1021 | 0.0964 | 0.1077 | 0.9616 |
| | **NICEM** | **280** | **13** | **0.0909** | **0.0892** | **0.0919** | **0.9721** |
| Cobalt | PLS | 400 | 12 | 0.0881 | 0.0851 | 0.0957 | 0.9649 |
| | MCUVE | 94 | 10 | 0.0786 | 0.0774 | 0.0838 | 0.9731 |
| | CARS | 45 | 8 | 0.0611 | 0.0549 | 0.0654 | 0.9836 |
| | GA-PLS | 76 | 8 | 0.0665 | 0.0614 | 0.0682 | 0.9822 |
| | NICE | 400 | 9 | 0.0534 | 0.0508 | 0.0571 | 0.9875 |
| | **NICEM** | **120** | **11** | **0.0341** | **0.0326** | **0.0356** | **0.9963** |
| Nickel | PLS | 400 | 12 | 0.3347 | 0.3042 | 0.3401 | 0.7339 |
| | MCUVE | 198 | 15 | 0.3442 | 0.3143 | 0.3562 | 0.7081 |
| | CARS | 74 | 13 | 0.3316 | 0.3272 | 0.3494 | 0.7155 |
| | GA-PLS | 91 | 14 | 0.3119 | 0.2915 | 0.3148 | 0.7719 |
| | NICE | 400 | 14 | 0.2131 | 0.2104 | 0.2188 | 0.8898 |
| | **NICEM** | **185** | **13** | **0.2019** | **0.2007** | **0.2028** | **0.9053** |
| Iron | PLS | 400 | 6 | 0.0392 | 0.0372 | 0.0408 | 0.9952 |
| | MCUVE | 122 | 10 | 0.0596 | 0.0562 | 0.0642 | 0.9881 |
| | CARS | 46 | 8 | 0.0339 | 0.0325 | 0.0359 | 0.9962 |
| | GA-PLS | 72 | 9 | 0.0319 | 0.0296 | 0.0324 | 0.9969 |
| | NICE | 400 | 13 | 0.0372 | 0.0362 | 0.0414 | 0.995 |
| | **NICEM** | **35** | **9** | **0.0205** | **0.0191** | **0.0217** | **0.9986** |

**Figure S5 Relationship between RMSECV and number of selected latent variables (LVs) in PLS and NICEM models for multi-metal ions datasets. (a) Zinc, (b) Copper, (c) Cobalt, (d) Nickel, (e) Iron.**

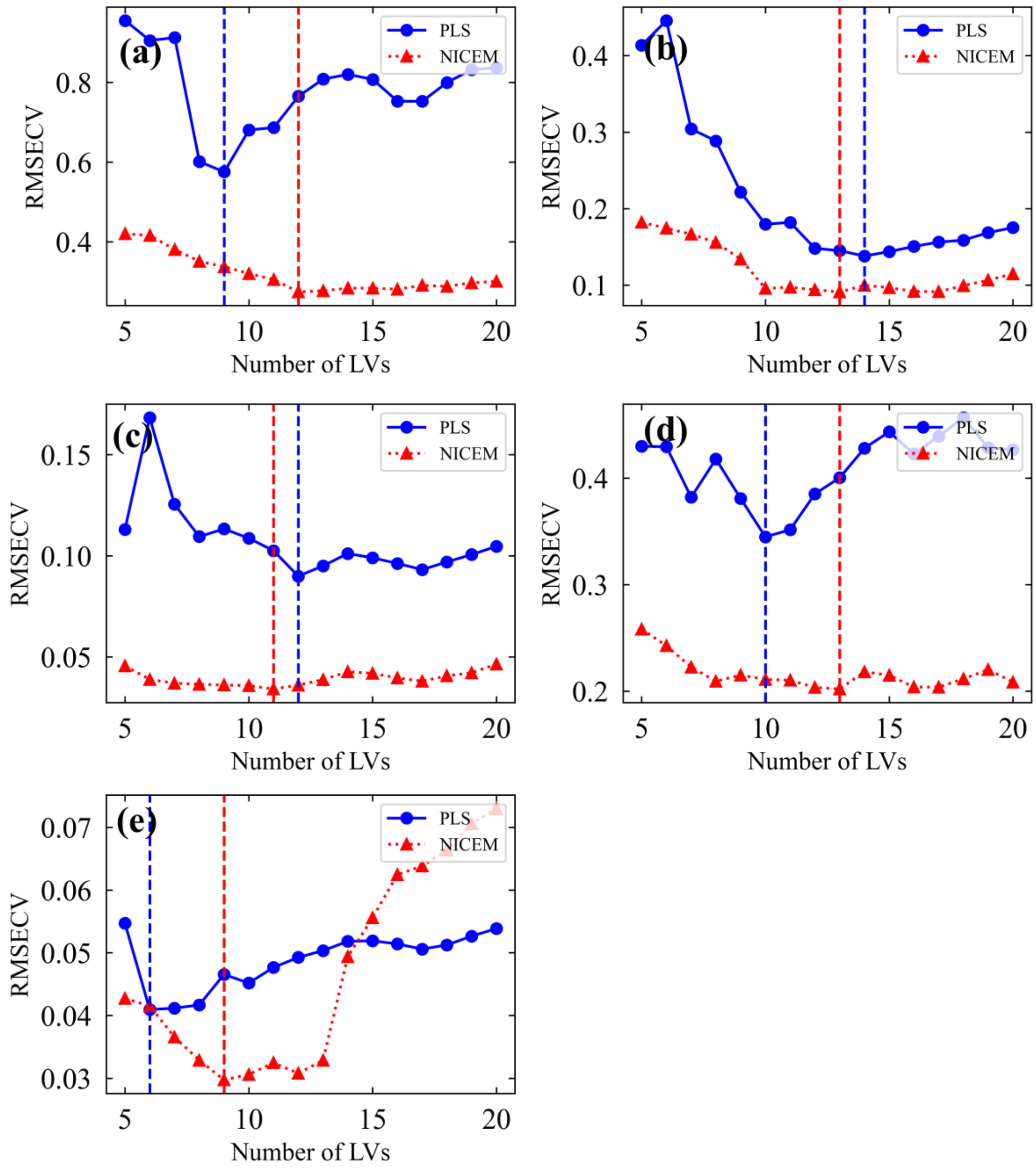
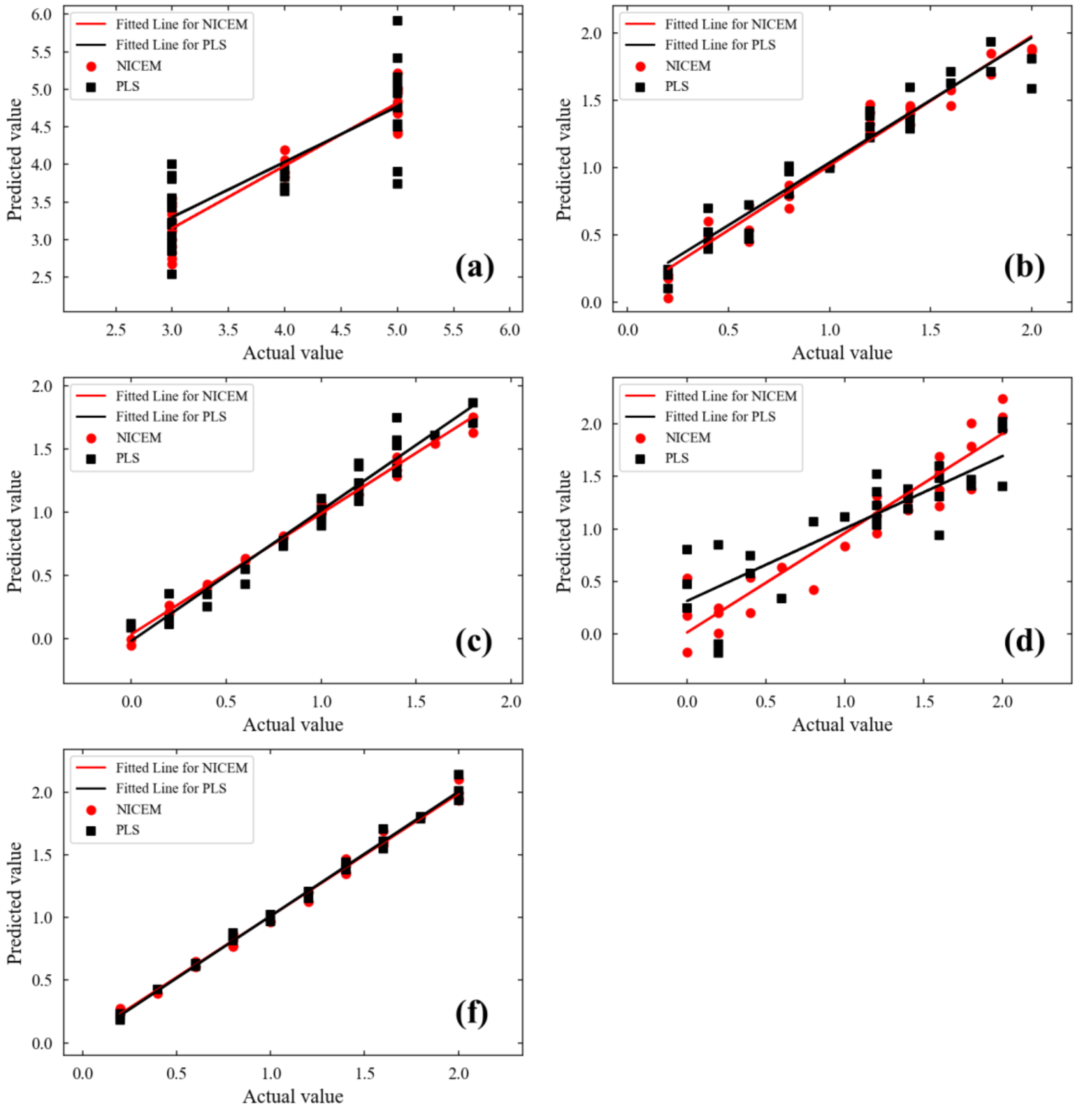


**Figure S6 Prediction results of PLS and NICEM models for multi-metal ions datasets. (a) Zinc, (b) Copper, (c) Cobalt, (d) Nickel, (e) Iron.**



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